Introduction

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Semester 2026-2

Welcome

"Build, Containerize and Deploy your first FOSS Service" Workshop

- **Audience:** Programmers with basic Linux knowledge.
- **Duration:** 6 to 8 hours.
- **Goal:** Build, containerize, and deploy a standard web service on Linux with secure HTTPS access.

Learning Objectives

1. Build a simple Flask application and containerize it using Podman.
2. Configure the minimum components of a production-ready Linux server for this service (firewall, systemd).
3. Deploy a containerized service with HTTPS using Nginx as a reverse proxy.

The Problem, The Goal, The Solution

Problem

Developers want to see their applications running in real conditions, but `localhost` falls short when they need to share a project or host a FOSS service.

Goal

Developers want a simple and secure way to make their applications available through a public internet URL.

Solution

In this workshop, participants will create, containerize, and deploy a basic Linux-hosted service with secure HTTPS access.

Workshop Materials

- A computer with an open ssh client.
- A VPS or VM with a Linux distribution (Ubuntu, Debian, Fedora, etc.).

Agenda

1. Workshop Introduction
2. App Building
3. Containerization
4. Server Setup
5. Reverse Proxy & TLS
6. Final Validation & Troubleshooting

Starting Validation

- Check your VPS/VM credentials.
- Ensure you have SSH access:

  ```
  ssh user@your-vps-ip
  ```

- Verify this repository is cloned locally and you have a text editor installed.